

MEC Defense: Securing Smart Stadium Critical Infrastructure

Team members

José Fernandes, nº114472

Diogo Ferreira, nº114002

Rúben Franco, nº114827

Diogo Carvalho, nº113221

Gonçalo Martins, nº112678

Tiago Rocha, nº120515

Supervisors

Prof. Daniel Corujo

Prof. Vitor Cunha



universidade
de aveiro



instituto de
telecomunicações





Our Motivation

Ease the log collecting process on MEC based systems

Provide a real-time status of the apps deployed on the system and the system itself

Facilitate the network owner/analyst side of the security process

Create a system capable of withstanding high volumes of network traffic



Current Progress

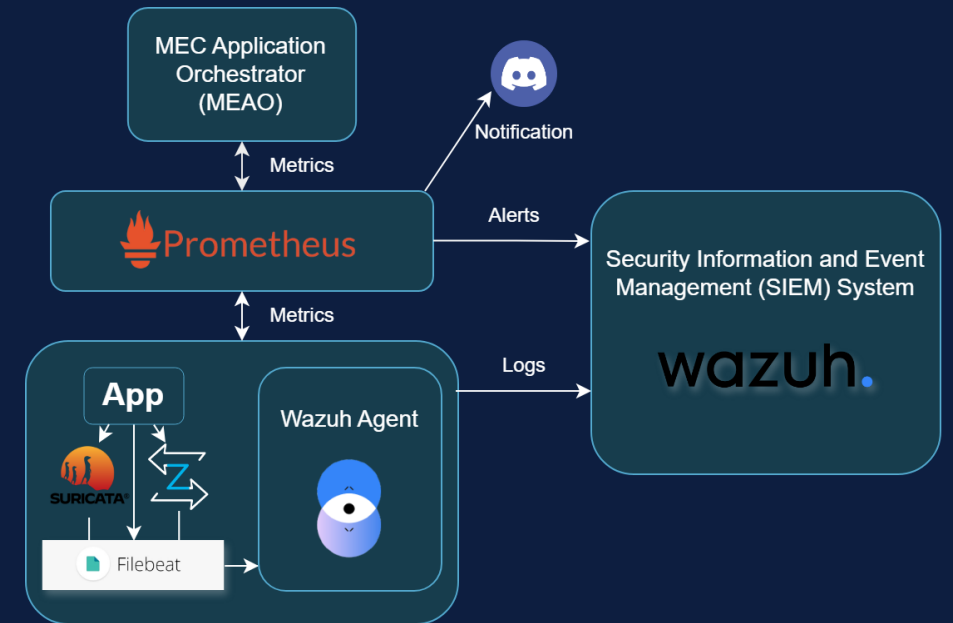
Logs being collected from the App's interface by Zeek and Suricata and sent to Filebeat

Established communication between the Wazuh agent and the Wazuh manager

Unique rulesets that prevent scenario specific events

Cluster metric collection and analysis with Prometheus

Video processing simulation app that follows standard log procedures





Future/Ongoing objectives

Report automation based on alerts for forensic analysis

External threat protection

Real-time attack mitigation

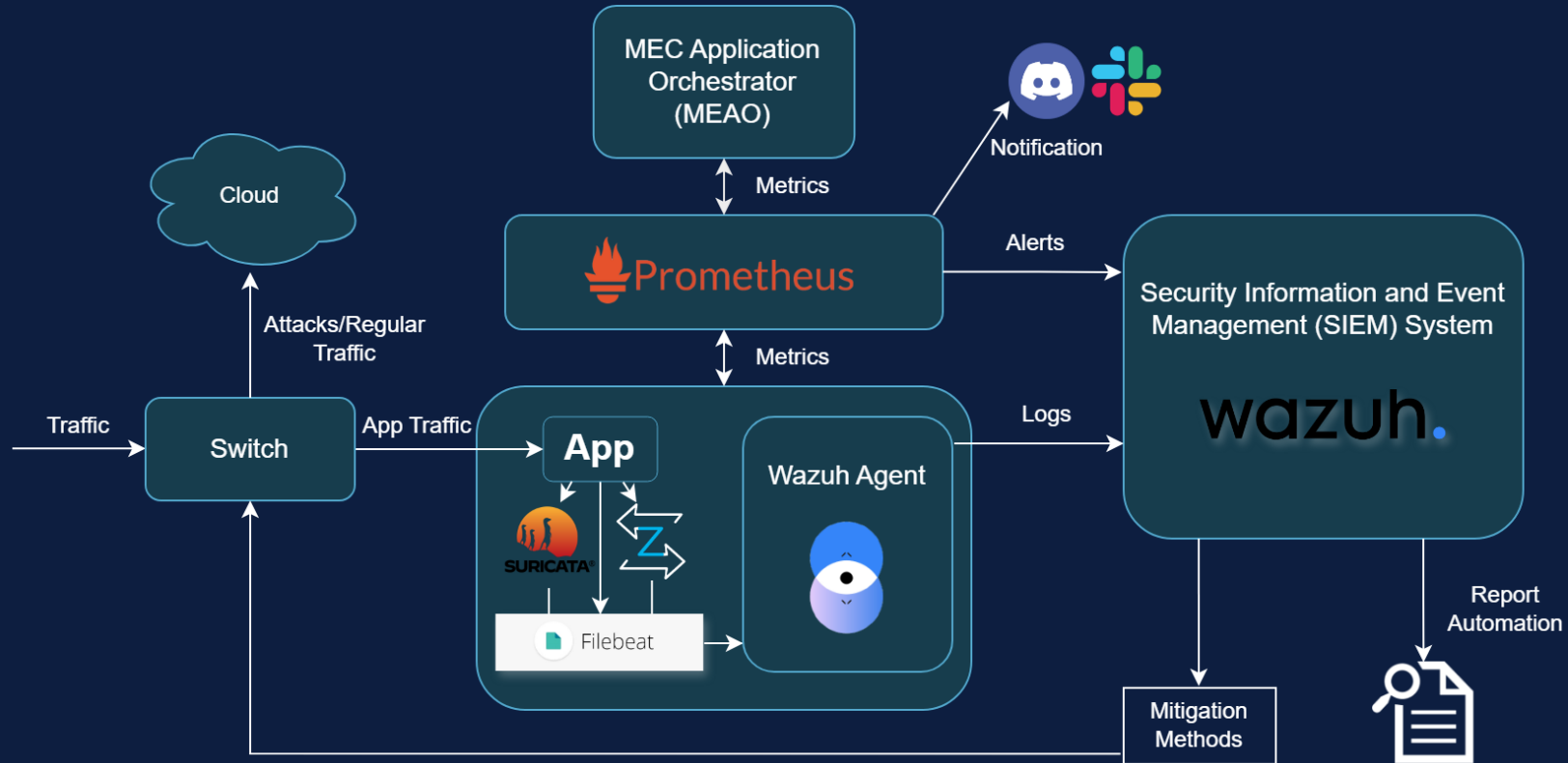
Network segmentation

Full system analysis

Redirection of ongoing attacks to a dedicated cloud infrastructure



Architecture



App – Prometheus – SIEM

Zeek – Filebeat – Agent – SIEM

Suricata– Filebeat – Agent – SIEM

App – Filebeat – Agent – SIEM



DEMO